

Module 1

Session 1

Q1. Name 5 IT job roles and write one line about what each role does.

Ans.

1. Developer – Develops and maintains software applications and websites.
2. Tester – Tests software to find bugs and checks whether it works correctly.
3. Designer – Creates the visual design and user interface of websites or applications.
4. Data Analyst – Collects, analyzes, and interprets data to help businesses make decisions.
5. AI Engineer – Develops and implements artificial intelligence and machine learning solutions.

Q 2. List 3 product companies and 3 service companies with one example of work each does.

Ans.

Product Companies:

1. Google – Develops products such as Google Search, Gmail, and Google Maps.
2. Zoho – Develops business software such as Zoho CRM and Zoho Books.

3. Microsoft – Develops products such as Windows, Microsoft 365, and Azure.

Service Companies:

1. TCS – Provides IT consulting, software development, and technology services to clients.
2. Infosys – Provides software development, consulting, and digital transformation services.
3. Wipro – Provides IT services, consulting, and business technology solutions.

Q 3. Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

Ans.

1. GitHub – GitHub is a platform used to store, manage, and collaborate on software code.
2. Slack – Slack is a communication and collaboration tool used by teams for messaging and file sharing.
3. Jira – Jira is a project management tool used to track tasks, issues, and software development work.
4. IDE – Integrated Development Environment; it is software used to write, run, and debug programs.
5. API – Application Programming Interface; it allows different software applications to communicate with each other.

Q 4. Name any 5 programming languages and mention one popular use case for each.

Ans.

1. Python – Used for web development, data analysis, and AI.
2. Java – Used for enterprise applications and Android development.
3. C++ – Used for system software, games, and performance-based applications.
4. JavaScript – Used to create interactive websites and web applications.
5. PHP – Used for server-side web development and dynamic websites.

Q 5. What is the difference between a developer and a tester?

Answer in 2 lines.

Ans.

Developer – Writes and develops software or applications and adds new features.

Tester – Tests the software to find bugs and checks whether it works according to requirements.

Session 2

Q 1. What are the three main layers of software architecture? Write one line about each.

Ans.

1. UI (User Interface) – This is the front-end layer through which users interact with the software.
2. Business Logic – This layer contains the rules and processes that control how the application works.
3. Database – This layer stores, retrieves, and manages the application's data.

Q 2. Define client and server in your own words with one real-life example.

Ans.

Client – A client is a device or application that sends a request for a service or data.

Server – A server is a computer or system that receives requests and provides the required service or data.

Example – In a web browser, the browser is the client and the website's server provides the requested webpage.

Q 3. Write 2 differences each between desktop apps, web apps, and mobile apps.

Ans.

Desktop Apps:

1. They are installed directly on a computer or laptop.
2. They usually depend on the computer's operating system and resources.

Web Apps:

1. They run through a web browser and usually do not need full installation.
2. They can be accessed from different devices using the internet.

Mobile Apps:

1. They are designed mainly for smartphones and tablets.
2. They are usually installed through app stores or app packages.

Q 4. Name any 3 databases and mention whether each is SQL or NoSQL.

Ans.

1. MySQL – SQL database.
2. PostgreSQL – SQL database.
3. MongoDB – NoSQL database.

Session 3

Q 1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Ans.

1. HTTP – HyperText Transfer Protocol; used to transfer web pages and data between a browser and a web server.
2. HTTPS – HyperText Transfer Protocol Secure; used to securely transfer web data using encryption.
3. FTP – File Transfer Protocol; used to transfer files between computers over a network.
4. SMTP – Simple Mail Transfer Protocol; used to send emails from one mail server or client to another.

Q 2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

Ans.

IPv4 uses 32-bit addresses and provides about 4.3 billion unique addresses.

IPv6 uses 128-bit addresses and provides a much larger number of unique addresses.

Q 3. Run nslookup google.com on your system and write down the IP address you get.

Ans.

The IP address returned by nslookup can vary depending on the DNS server and location. A sample IPv4 address is 142.250.191.14.

Note: Run the command `nslookup google.com` on your system to write the exact IP address shown on your computer.

Q 4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

Ans.

200 – OK: The request was successful.

301 – Moved Permanently: The requested resource has been permanently moved to a new URL.

404 – Not Found: The requested resource could not be found on the server.

500 – Internal Server Error: The server encountered an unexpected problem.

403 – Forbidden: The server understood the request but refuses to allow access.

Q 5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

Ans.

Root (.) – The top level of the DNS hierarchy. It directs queries to the appropriate TLD servers.

TLD (.com) – The Top-Level Domain in www.google.com is .com.

Domain (google) – 'google' is the main domain name registered under the .com TLD.

Subdomain (www) – 'www' is a subdomain/host name used before the main domain.

Example: www.google.com → Root (.) → TLD (.com) → Domain (google) → Subdomain/Host (www).

Session 4

Q 1. List all 6 phases of SDLC in order with one line about each.

Ans.

Requirement – Collect and understand the user's and business requirements.

Design – Plan the software structure, user interface, database, and system architecture.

Development – Developers write the code and build the software.

Testing – Testers check the software for bugs and verify that requirements are met.

Deployment – The completed software is released and made available to users.

Maintenance – The software is updated, fixed, and improved after release.

Q 2. Write 3 differences between Agile and Waterfall models.

Ans.

Agile develops software in small, repeated cycles, while Waterfall follows a fixed step-by-step process.

Agile allows requirements to change during development, while Waterfall generally expects requirements to be defined early.

Agile delivers working software frequently, while Waterfall usually delivers the complete product after the development process is finished.

Q 3. Name the 3 main Scrum roles and write one line about the responsibility of each.

Ans.

Product Owner (PO) – Manages product requirements and prioritizes the work for the team.

Scrum Master – Helps the team follow Scrum practices and removes obstacles affecting the team.

Development Team – Designs, develops, tests, and delivers the product increment.

Q 4. What is a Sprint in Scrum? Mention its typical duration.

Ans.

A Sprint is a fixed period in Scrum during which the team works to complete a planned set of work and create a usable product increment.

Typical duration: A Sprint usually lasts 1 to 4 weeks, with 2 weeks being common.

Q 5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

Ans.

To Do – Tasks that are planned but have not been started yet.

In Progress – Tasks that the team is currently working on.

Done – Tasks that have been completed and meet the required conditions.

Backlog – A list of planned or pending work items that may be worked on in the future.

Session 5

Q 1. Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes.

Ans.

Initialize a repository: `git init` Stage all files: `git add .`

Commit with a message: `git commit -m "Initial commit"` Push to remote: `git push origin main` Pull latest changes: `git pull origin main`

Q 2. What is the difference between git fetch and git pull?

Answer in 2 lines.

Ans.

`git fetch` downloads the latest changes from the remote repository but does not merge them into the current branch. `git pull` downloads the latest changes and automatically merges them into the current branch.

Q 3. Create a new GitHub repository, push any small text file from your local system, and share the repo link.

Ans.

Create a new repository on GitHub, connect your local repository to it, and push the text file. Commands: `git init` → `git add .` → `git commit -m "Initial commit"` → `git branch -M main` → `git remote add origin` → `git push -u origin main` Repository Link.

Q 4. What is the difference between a Fork and a Pull Request?

Answer in 2 lines.

Ans.

Fork – Creates a personal copy of another user's GitHub repository under your own GitHub account. Pull Request – A request to merge changes from one branch or repository into another repository's branch.

Q 5. Create a new branch named feature-test, make one commit on it, and merge it back to the main branch. Write the commands you used.

Ans.

Create and switch to the branch: `git checkout -b feature-test`
Stage the changes: `git add .` Make a commit: `git commit -m "Add feature-test changes"` Switch to main branch: `git checkout main`
Merge the branch: `git merge feature-test`.